

1. Introduction to Python

Python is a simple and powerful programming language used in:

- Data Analysis
- Machine Learning
- Web Development
- Automation

Features

- Easy to learn
 - Easy syntax
 - Open source
-

2. Variables

Variables are used to store data.

Example:

```
name = "Sachin"  
age = 21
```

3. Data Types

Data Type Example

Integer 10

Float 8.5

String "Sachin"

Boolean True

4. Input and Output

Input

```
name = input("Enter your name: ")
```

Output

```
print(name)
```

5. Loops

Loops are used to repeat tasks.

For Loop

```
for i in range(5):  
    print(i)
```

While Loop

```
count = 1  
while count <= 5:  
    print(count)  
    count += 1
```

6. Functions

Functions help reuse code.

```
def add(a, b):  
    return a + b  
  
print(add(10, 20))
```

7. Lists

Lists store multiple values.

```
numbers = [10, 20, 30, 40]
```

Common operations:

```
numbers.append(50)  
len(numbers)
```

8. Jupyter Notebook

Jupyter Notebook is used for:

- Writing Python code
- Data Analysis
- Creating reports

Today's Achievement

Installed Jupyter Notebook

9. Internship Learning

Today I learned:

- Python Basics
- Variables
- Data Types
- Loops
- Functions
- Lists
- Jupyter Notebook Setup